For this midterm, you are allowed to use your textbook (Hartle) and your personal notes. However, please refrain from using other sources, including the internet. An exception can be made for the starred ($\star$) question. The questions marked with a ($\star$) are required for students enrolled in 583 and bonus for students enrolled in 483.

1. Consider a 2-sphere with coordinates (θ, ϕ) and metric

$$ds^2 = d\theta^2 + \sin^2\theta d\phi^2 \,. \tag{1}$$

(a) Show that lines of constant longitude (ϕ =constant) are geodesics, and that the only line of constant latitude (θ =constant) that is a geodesic is the equator ($\theta = \pi/2$).

Solution: The nonzero components of the metric and inverse metric are

$$g_{\theta\theta} = 1, \qquad g_{\phi\phi} = \sin^2\theta,$$
$$g^{\theta\theta} = 1, \qquad g^{\phi\phi} = \sin^{-2}\theta.$$

The nonvanishing Christoffel symbols are

$$\Gamma^{\phi}_{\phi\theta} = \Gamma^{\phi}_{\theta\phi} = \frac{1}{2}g^{\phi\phi}\partial_\theta g_{\phi\phi} = \frac{\cos\theta}{\sin\theta},$$
$$\Gamma^{\theta}_{\phi\phi} = -\frac{1}{2}\partial_\theta g_{\phi\phi} = -\sin\theta\cos\theta.$$

Constant longitude. Parameterize the curve as

$$x^\mu(\lambda) = (\lambda, \phi_0).$$

The ϕ geodesic equation is

$$\frac{d^2\phi}{d\lambda^2} + 2\Gamma^{\phi}_{\phi\theta}\frac{d\phi}{d\lambda}\frac{d\theta}{d\lambda} = 0,$$

which is satisfied since $d\phi/d\lambda = 0$. The θ equation becomes

$$\frac{d^2\theta}{d\lambda^2} - \sin\theta\cos\theta\left(\frac{d\phi}{d\lambda}\right)^2 = 0,$$

which is also satisfied. Thus lines of constant longitude are geodesics.

Constant latitude. Parameterize

$$x^\mu(\lambda) = (\theta_0, \lambda).$$

The θ equation gives

$$-\sin\theta_0\cos\theta_0 = 0,$$

which is satisfied only for $\theta_0 = \pi/2$ (the equator) or at the poles. Thus only the equator is a nontrivial geodesic of constant latitude.

(b) Take a vector with components $V_\mu = (1, 0)$ and parallel-transport it once around a circle of constant latitude. What are the components of the resulting vector, as a function of θ?

Solution: The parallel transport equation is

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu_{\sigma\rho}\frac{dx^\sigma}{d\lambda}V^\rho = 0.$$

Along the path $x^\mu = (\theta, \lambda)$, the equations become

$$\begin{aligned}\frac{dV^\theta}{d\lambda} &= \sin\theta\cos\theta\, V^\phi,\\ \frac{dV^\phi}{d\lambda} &= -\frac{\cos\theta}{\sin\theta}\, V^\theta.\end{aligned}$$

Combining,

$$\frac{d^2V^\theta}{d\lambda^2} + \cos^2\theta\, V^\theta = 0,$$

with solution

$$V^\theta(\lambda) = \cos(\omega\lambda), \qquad V^\phi(\lambda) = -\frac{1}{\sin\theta}\sin(\omega\lambda), \quad \omega = \cos\theta.$$

After one full revolution,

$$V^\mu(2\pi) = \left(\cos(2\pi\cos\theta), -\frac{\sin(2\pi\cos\theta)}{\sin\theta}\right).$$

The vector returns to itself only at the equator.

(c) Consider the following metric for one space-like and one time-like dimension:

$$ds^2 = t^{-2}(-dt^2 + dx^2)\,. \tag{2}$$

Find all the Christoffel symbols, and then calculate the geodesics of this spacetime. *Hint: for this second part, it is easiest to start with the definition of a geodesic as a maximum of proper time and find a relation for x and t by calculus of variations.*

Solution: The metric and inverse metric components are

$$g_{tt} = -t^{-2}, \quad g_{xx} = t^{-2}, \qquad g^{tt} = -t^2, \quad g^{xx} = t^2.$$

The nonzero Christoffel symbols are

$$\begin{aligned}&\Gamma^t_{tt} = -t^{-1}, && \Gamma^t_{xx} = -t^{-1},\\ &\Gamma^x_{tx} = \Gamma^x_{xt} = -t^{-1}.\end{aligned}$$

The proper time functional is

$$\int \sqrt{-ds^2} = \int \frac{\sqrt{1-\dot{x}^2}}{t}\, dt, \qquad \dot{x} = \frac{dx}{dt}.$$

Applying the Euler–Lagrange equation gives

$$\frac{d}{dt}\left(\frac{\dot{x}}{t\sqrt{1-\dot{x}^2}}\right) = 0,$$

so

$$\frac{\dot{x}}{\sqrt{1-\dot{x}^2}} = At.$$

Solving,

$$\dot{x} = \frac{At}{\sqrt{1+A^2t^2}},$$

which integrates to

$$x(t) = \sqrt{t^2 + A^{-2}} = \sqrt{t^2 + t_0^2}.$$

2. Consider the following line element,

$$ds^2 = -\left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right) dt^2 + \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)^{-1} dr^2 + (r^2+b^2)(d\theta^2 + \sin^2\theta d\phi^2)\,, \quad (3)$$

where M and b are positive, real constants. Note that we are using geometrical units where $G = c = 1$.

(a) Write down the metric and the inverse metric in the (t, r, θ, ϕ) coordinate system.

(b) Write down the non-zero Christoffel symbols $\Gamma^\alpha_{\beta\gamma}$ for this metric. *Hint: although there are 40 Christoffel symbols that are distinct by taking account of the symmetry under $\beta \leftrightarrow \gamma$, only 9 are non-zero.*

Solution: To do the calculation by hand, one can note that the non-zero metric derivatives are

$$g_{tt,r} = -f_{,r},\ g_{rr,r} = -\frac{f_{,r}}{f^2},\ g_{\theta\theta,r} = 2r,\ g_{\phi\phi,r} = 2r\sin^2\theta,\ g_{\phi\phi,\theta} = 2r^2\sin\theta\cos\theta$$

where I have defined $f(r) = \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)$. Thus, $f_{,r} = 2Mr(r^2+b^2)^{-3/2}$.

From these expressions, we can compute the Christoffel symbols to see that

$$\Gamma^t_{tr} = \Gamma^t_{rt} = \frac{f_{,r}}{2f}, \tag{4}$$

$$\Gamma^r_{tt} = \frac{1}{2} f f_{,r}, \tag{5}$$

$$\Gamma^r_{rr} = -\frac{f_{,r}}{2f}, \tag{6}$$

$$\Gamma^\theta_{\theta r} = \Gamma^\theta_{r\theta} = \frac{r}{b^2 + r^2}, \tag{7}$$

$$\Gamma^r_{\theta\theta} = -rf, \tag{8}$$

$$\Gamma^\phi_{\phi r} = \Gamma^\phi_{r\phi} = \frac{r}{b^2 + r^2}, \tag{9}$$

$$\Gamma^r_{\phi\phi} = -rf \sin^2\theta, \tag{10}$$

$$\Gamma^\phi_{\phi\theta} = \Gamma^\phi_{\theta\phi} = \frac{\cos\theta}{\sin\theta}, \tag{11}$$

$$\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta \tag{12}$$

(c) Write the geodesic equation in this metric for a massive particle, i.e. write equations for $\ddot{t}, \ddot{r}, \ddot{\theta}, \ddot{\phi}$ where the double-dots are with respect to proper time τ.

Solution: The geodesic equations are given by $\ddot{x}^\alpha = -\Gamma^\alpha_{\beta\gamma}\dot{x}^\beta\dot{x}^\gamma$.
Then, we have

$$\ddot{t} = -\frac{f_{,r}}{f}\dot{t}\dot{r}, \tag{13}$$

$$\ddot{r} = -\frac{1}{2} f f_{,r}\dot{t}^2 + \frac{f,r}{2f}\dot{r}^2 + rt(\dot{\theta}^2 + \sin^2\theta\dot{\phi}^2) \tag{14}$$

$$\ddot{\theta} = -\frac{2r}{b^2 + r^2}\dot{r}\dot{\theta} + \sin\theta\cos\theta\dot{\phi}^2, \tag{15}$$

$$\ddot{\phi} = -\frac{2r}{b^2 + r^2}\dot{r}\dot{\phi} - 2\frac{\cos\theta}{\sin\theta}\dot{\theta}\dot{\phi}. \tag{16}$$

(d) Calculate the effective potential for timelike orbits for arbitrary M and b.

Solution: To get the effective potential, we note that the metric is independent of t and ϕ, so there are conserved quantities e and l similar to the case of the Schwarzschild metric. WLOG, let's look at orbits in the $\theta = \pi/2$. The analogues here are

$$e = \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)\frac{dt}{d\tau} \ell = (r^2+b^2)\frac{d\phi}{d\tau}$$

the normalization of the 4-velocity $u^\alpha u_\alpha = -1$ gives us

$$-1 = -\left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)(u^t)^2 + \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)^{-1}(u^r)^2 + (r^2+b^2)(u^\phi)^2 \tag{17}$$

$$= -\left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)^{-1} e^2 + \left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right)^{-1}(u^r)^2 + \frac{\ell^2}{r^2+b^2} \tag{18}$$

After some manipulation, we see that the effective potential for timelike orbits is given by

$$V_{\text{eff}} = \frac{\ell^2}{2}\frac{1}{r^2+b^2}\left(1 - \frac{2M}{\sqrt{r^2+b^2}}\right) - \frac{M}{\sqrt{r^2+b^2}}$$

(e) Do bound orbital solutions exist for $M = 0$? Why or why not?

Solution: To calculate the bound circular orbits for $M \to 0$, we solve $\partial V_{\text{eff}}/\partial r = 0$ and find that only $r = 0$ satisfies this equation. Therefore, there are no circular orbits.

(f) ($\star$) Build a numerical solver for integrating the geodesic equation given an initial position ϵ and ℓ. Here, ϵ corresponds to energy density as written in class. Can you find any interesting geometries and/or trajectories?